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Kaveh Karimadini & Sharareh Pourebrahim

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Simulation of Long-Term Spatio-Temporal Environmental Dynamics using a Unified Benchmark of Neighbor Augmenting, LSTM and Graph Attention Models

Kaveh Karimadini¹, Sharareh Pourebrahim^{2, 3*}

- 1- Master student, School of Computing, Macquarie University, Sydney, Australia, kavehkarimoodini@gmail.com, kaveh.karimadini@students.mq.edu.au
- 2- Jeffrey Sachs Center on Sustainable Development, School of Business, Sunway University, 47500 Bandar Sunway, Petaling Jaya, Malaysia, shararehp@sunway.edu.my
- 3- Sunway Institute for Global Strategy & Competitiveness, Sunway University, Bandar Sunway, 47500, Selangor, Malaysia

Abstract

Rapid environmental change has increased the need for predicting the long-term geospatial reliably. However, accurately modeling spatio-temporal geospatial dynamics remains challenging. Because of the nonlinearities, complex spatial dependency, and external driving factors, it is difficult to predict. In this paper, a comprehensive benchmarking framework is proposed for the comparison of neighborhood-based, graph-based and attention-based spatiotemporal deep learning models, with the same preprocessing, training and testing procedure. Long Short-Term Memory (LSTM) models with and without auxiliary variables are compared with hybrid Graph Attention Network–LSTM (GAT–LSTM) models and fully attention-based GAT–Temporal Attention models, with and without a feed-forward (MLP/FFN) block. All models are trained using a unified preprocessing and evaluation pipeline on annual satellite data in Network Common Data Form (NetCDF) from 2000 to 2023, with 2024 reserved as a fully unseen test dataset. Global pixel-wise measures such as R^2 , RMSE, MAE, MAPE, and correlation are used to evaluate model performance based on performance of vectors and alignment of predicted vectors and reference vectors. Findings indicate that the LSTM–CA with auxiliary inputs (3×3 neighborhood) performs the best and most stable performance ($R^2 \approx 0.95$), highlighting the importance of the integrated Cellular Automata (CA) structure and auxiliary driving factors. The GAT–Temporal Attention model with an MLP block ranks second, while removing the MLP or using hybrid LSTM–GAT configurations lead to unstable or degraded performance. Index-wise analysis shows that vegetation and water-related indices are more predictable. The results indicate that strong temporal modeling of information combined with auxiliary information is more important than complexity of spatial attention. The main novelty of this paper is that it does not introduce a new model for a neural network, instead it proposes a comparative engineering experiment to assess the conditions where neighborhood-based temporal models could be superior to graph-attention models in geospatial long-range prediction applications.

Keywords:

Spatio-temporal forecasting; Geospatial time series; LSTM; Graph Attention Network (GAT); Environmental monitoring.

1. Introduction

Spatio-temporal prediction plays a central role in a wide range of applications, including transportation systems (Wang et al., 2025), air pollution and weather monitoring (Tan et al., 2022; Guo et al., 2026), land-use planning (Wang et al., 2023; Liu et al., 2021), agricultural productivity assessment (Anikó Nyéki et al., 2021), smart city and urban development (Guangyin et al., 2024), environmental management (Rezaie et al., 2024), and public health (Tao et al., 2021; Yudong et al., 2022). In these domains, reliable forecasting supports proactive planning, risk mitigation, and more efficient allocation of resources (Kumar et al., 2024). However, learning the joint evolution of data across space and time remains a challenging task due to the complex, nonlinear, and non-stationary nature of real-world processes, as well as the rapidly growing volume and heterogeneity of data generated by sensors and satellite observations. As noted by Kumar et al. (2024), joint spatio-temporal prediction requires modeling tightly coupled dependencies across both dimensions while handling high-dimensional and heterogeneous data sources.

Traditional probabilistic and shallow machine learning models often struggle to capture such complexities, particularly when faced with strong nonlinearities, non-stationarity, and long-range dependencies. These limitations have motivated the increasing adoption of deep learning approaches, which offer greater representational capacity and flexibility for modeling complex spatio-temporal dynamics.

1.2 Advances and Limitations in Recent Spatio-Temporal Models

Since around 2021, deep learning has become a dominant paradigm in spatio-temporal forecasting. Hybrid architectures that combine Graph Neural Networks (GNNs) with sequential models such as Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) (Smagulova et al., 2019) networks have gained popularity, especially in traffic and mobility applications. For example, Simini et al. (2021) introduced the Deep Gravity model for predicting human mobility flows by leveraging geographic features and deep neural networks, achieving improved accuracy over classical gravity models and better generalization to data-scarce regions. In parallel, Transformer-based models have had a major impact on time-series modeling. The Informer model (Zhou et al., 2021) introduced a sparse self-attention mechanism to address the quadratic complexity of standard Transformers, enabling the modeling of long-range temporal dependencies in large-scale datasets. These ideas have since been extended to spatio-temporal contexts, including air quality forecasting (Liang et al., 2023) and global weather prediction (Pathak et al., 2022; Lam et al., 2023).

Despite these advances, important limitations remain. [Zhou et al. \(2024\)](#) reported that fully attention-based spatio-temporal models, such as GMAN ([Zheng et al., 2020](#)), can produce overly diffuse and uninformative attention maps, suggesting that explicit spatial structure must be incorporated into model architectures. At the same time, large-scale models such as FourCastNet ([Pathak et al., 2022](#)) and GraphCast ([Lam et al., 2023](#)) have demonstrated the remarkable potential of data-driven approaches for high-resolution weather forecasting, but they also highlight the growing complexity and computational demands of modern spatio-temporal models. [Pathak et al. \(2022\)](#) unveiled FourCastNet, a global forecasting model using adaptive Fourier neural operators. FourCastNet matched the precision of the European Centre’s state-of-the-art numerical weather prediction model over small scales and even surpassed it in more complex phenomena like precipitation, in about under 2 seconds of real-time data, making the week-long prediction that took five orders of magnitude more time using conventional models. The data-driven emulator demonstrated how AI models can be more efficient. For example, by the year 2023, spatio-temporal prediction technologies had branched out into different scales of applicability, with some models utilizing the hybrid architecture of transformers and GNNs. In air quality prediction, for instance, the study by [Liang et al. 2023](#) proposed a novel air quality prediction model called AirFormer, characterized by its two-stage transformer architecture. The first stage consists of two self-attention modules for effective spatio-temporal feature learning. The model also incorporates latent variables for addressing uncertainty associated with air pollution data. With data sampled over 4 years and more than 1,000 monitoring stations, this model realizes a 5-8 % prediction error reduction compared to the state-of-the-art. Concurrently, [Lam et al. 2023](#) took weather forecasting further with the GraphCast model, a multi-scale graph neural network model used in forecasting the weather. GraphCast is characterized as a model of the global atmosphere as a graph where it learns to predict hundreds of weather variables looking ahead 10 days. Notably, it has beaten the world’s best 10-day forecast system in 90% of 1,380 verification metrics, particularly in forecasting phenomena such as tropical cyclones or heatwaves. The main benefit of Graph Attention Network (GAT) ([Velićković et al., 2017](#)) over Graph Convolutional Network (GCN) is that GAT is anisotropic, meaning it learns edge-specific attention weights and can assign different importance to different neighbors. In contrast, GCN is isotropic, applying the same transformation uniformly across neighbors ([Chen, & Yuan, 2023](#)), which limits its ability to model asymmetric or context-dependent relationships.

Emerging trends also suggest that spatial-temporal forecasting is increasingly adopting specialized architectural approaches for multi-scale dependency learning, missing data resilience, and non-stationary temporal dynamics. For instance, MGSFformer ([Yu et al., 2025](#)) implements multi-granularity spatio-temporal fusion in air quality prediction, underscoring the relevance of

modeling dependencies across multiple spatial and temporal resolutions. On the other hand, GinAR+ (Yu et al., 2025) solves the problem of multivariate time-series forecasting in scenarios involving missing values, which is particularly pertinent in remote sensing, given the frequent presence of clouds that may obstruct satellite images. Finally, MetaGNSDformer (Cheng et al., 2025) integrates meta-learning, gated non-stationary dynamics, and frequency-aware attention to solve point-interval prediction, indicating an emerging focus on models capable of handling distribution shift and non-stationary temporal dynamics. While these papers differ in their application areas, they all emphasize the significance of benchmarking tools that can test models on accuracy, robustness, missing data handling, and generalization to novel environmental conditions. Lastly, heterogeneous ensemble surrogate-assisted optimization techniques, as seen in lithium-ion battery charging strategy optimization, are part of the wider trend of integrating different model families to enhance robustness against nonlinear system dynamics (Cheng & Liu, 2025).

1.3 Hybrid Architectures for Spatio-Temporal Modeling: Progress and Limitations

Recent years have seen increasing interest in hybrid architectures that integrate signal processing, graph-based learning, and recurrent or attention-based temporal modeling. For instance, Wang et al. (2024) proposed a VMD–GAT–BiLSTM model, combining Variational Mode Decomposition (VMD), Graph Attention Networks (GAT), and Bidirectional LSTM (BiLSTM), for urban air quality forecasting, achieving substantial reductions in MAE and notable improvements in R^2 across multiple cities. In traffic forecasting, Ma et al. (2024) introduced the spatio-temporal traffic large model (STTLM), demonstrating the benefits of Transformer-based pretraining with contextual information.

Beyond conventional sensor data, novel modalities have also been explored. Xu et al. (2025) proposed Imagery2Flow, which predicts human mobility flows directly from satellite imagery using a combination of computer vision and graph learning. In the seismic domain, Leema et al. (2025) introduced SeismoQuakeGNN, a hybrid GNN–Transformer framework that achieved an R^2 of 0.85 for earthquake magnitude prediction, illustrating the growing maturity of hybrid spatio-temporal modeling approaches.

In the specific context of land use/cover change (LUCC) modeling, hybrid approaches combining Cellular Automata (CA) and LSTM have shown promising results. Rezaie et al. (2024) demonstrated that an LSTM–CA model with class weighting can effectively address data imbalance and achieve high classification accuracy, precision, recall, and F1-score, significantly outperforming traditional Markov-based methods. This highlights the complementary roles of CA in representing local spatial neighborhood interactions and LSTM

in capturing temporally informed transition rules. More recently, [Li et al. \(2024\)](#) explored the integration of Transformers with CA for dynamic LUCC simulation, pointing to the underexplored potential of attention mechanisms in this domain. [Fan et al. \(2025\)](#) further proposed ExHAGN, a dynamic heterogeneous graph framework that models human activities and land-use changes jointly, achieving consistent improvements over state-of-the-art statistical and deep learning baselines.

However, Spatio-temporal modeling is inherently affected by both model-form and parametric uncertainties. The nonlinear and non-stationary nature of geospatial processes makes the selection of an optimal architecture difficult, and different modeling assumptions or hyperparameter choices can lead to substantially different predictions. This often results in ensemble divergence, where multiple models produce conflicting outcomes for the same scenario.

Although a wide range of statistical, machine learning, and deep learning techniques have been proposed, most studies focus on introducing new architecture rather than systematically comparing fundamentally different modeling paradigms under a common evaluation framework, particularly for long-term generalization to a fully unseen future time step.

1.4 Research Gap and Motivation

Two main gaps can be identified in current literature. First, there is a lack of engineering-level comparative studies that evaluate neighborhood-based models such as neighborhood augmented LSTM against graph-based attention models under a unified data preprocessing, training, and testing pipeline. Second, most existing studies emphasize short-term accuracy or application-specific performance, while long-term generalization to an entirely unseen future year remains underexplored. This limitation is particularly critical for engineering applications, where forecasts are intended to support forward-looking planning and decision-making.

Addressing these gaps requires evaluation frameworks that (i) explicitly capture both local spatial interactions and long-range spatial dependencies, (ii) model temporal dynamics using recurrence and/or attention mechanisms, and (iii) assess predictive performance, under (i) and (ii), on a truly unseen future period. This research paper describes comparative experiments carried out under controlled conditions, as opposed to research work which is predominantly aimed at the design of new architecture. The novelty of this work is that different spatio-temporal modelling strategies are applied and experimented under similar conditions: NA-LSTM (Neighborhood Augmented Long Short Term Memory), Graph Attention Recurrent Model, Fully Attention-Based Temporal Model.

1.5 Objective and Role of the Proposed Methods

Motivated by these considerations, the main objective of this study is to provide a controlled benchmarking framework that systematically compares neighborhood-augmented LSTM forecasting and graph-attention-based architectures for long-term spatio-temporal geospatial forecasting. (1-1 and Reviewer 3-2,1) The core task is the one-year-ahead prediction of multi-layer geospatial states using historical data from 2000 to 2023, with the transition from 2023 to 2024 reserved exclusively for testing. Three complementary modeling approaches are evaluated:

1. neighborhood-augmented LSTM, which encodes local spatial neighborhood interactions through a neighborhood augmentation formulation and learns temporally informed transition rules using LSTM networks.
2. GAT-LSTM, which applies Graph Attention Networks to capture spatial dependencies followed by LSTM-based temporal modeling.
3. GAT with Temporal Attention, which replaces recurrent temporal modeling with Transformer-style self-attention to capture long-range temporal dependencies in a fully attention-based manner.

By evaluating these architectures under a common data preprocessing pipeline and a consistent long-term forecasting setup, this study provides an engineering-focused comparison of neighborhood-based, graph-based, recurrent, or attention-driven spatio-temporal models. The results aim to clarify their respective strengths, limitations, and generalization behavior, and to offer practical guidance for the design of robust spatio-temporal forecasting systems in geospatial and environmental engineering applications.

2. Materials and Methods

The proposed framework comprises (i) harmonized spatio-temporal data preprocessing, (ii) spatial feature construction using either cellular automata (CA) neighborhood encoding or graph attention mechanisms, (iii) temporal modeling via LSTM or multi-head self-attention, (iv) supervised one-step-ahead training with the Huber loss, and (v) post-processing and vectorized multi-band evaluation. This unified pipeline ensures a consistent and fair comparison across all model families. [Figure 1](#) illustrates the overall methodological framework of the study.

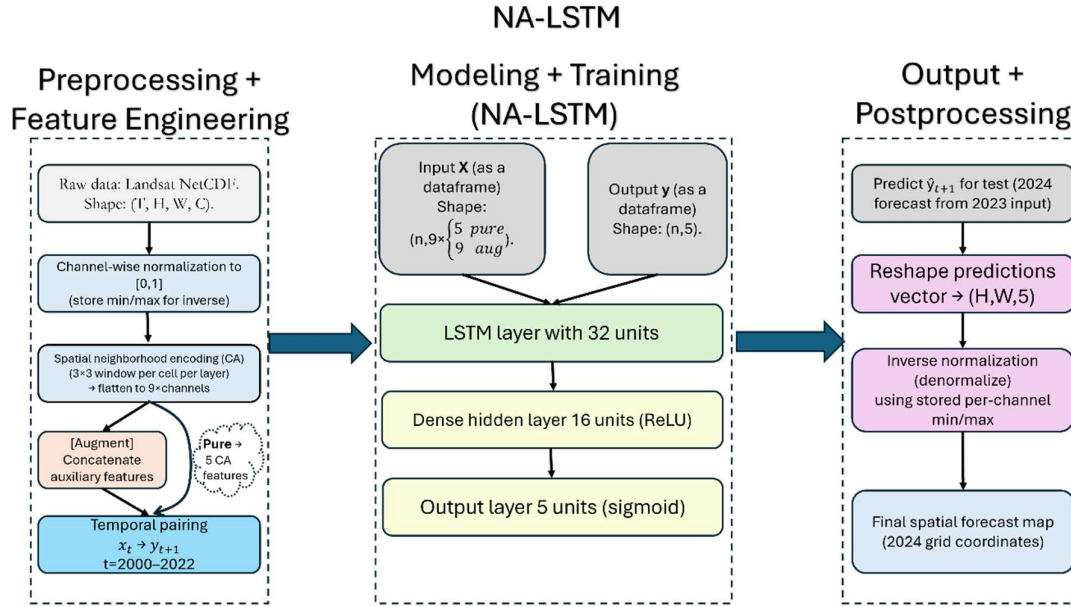


Figure 1a. The full pipeline of the NA-LSTM modeling

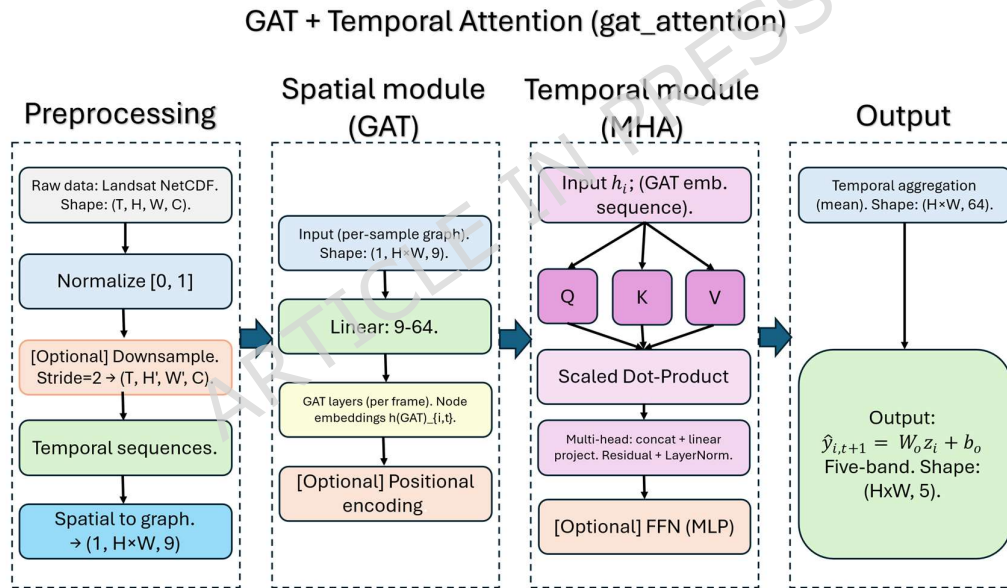


Figure 1b. The full pipeline of the GAT-Temporal Attention modelling

2.1 Study Area

The study was conducted in the Klang River Basin, located in the western region of Peninsular Malaysia (Figure 2). The basin covers approximately 127,326 hectares and includes major parts of the Klang Valley, including Kuala Lumpur and large areas of Selangor. As Malaysia's main administrative, commercial, and financial region, the basin is highly urbanized and undergone rapid land use changes over recent decades. These changes have altered hydrological processes, surface temperature patterns, ecosystem services, and overall environmental quality, making the

basin a suitable testbed for long-term spatio-temporal forecasting under strong anthropogenic pressure.

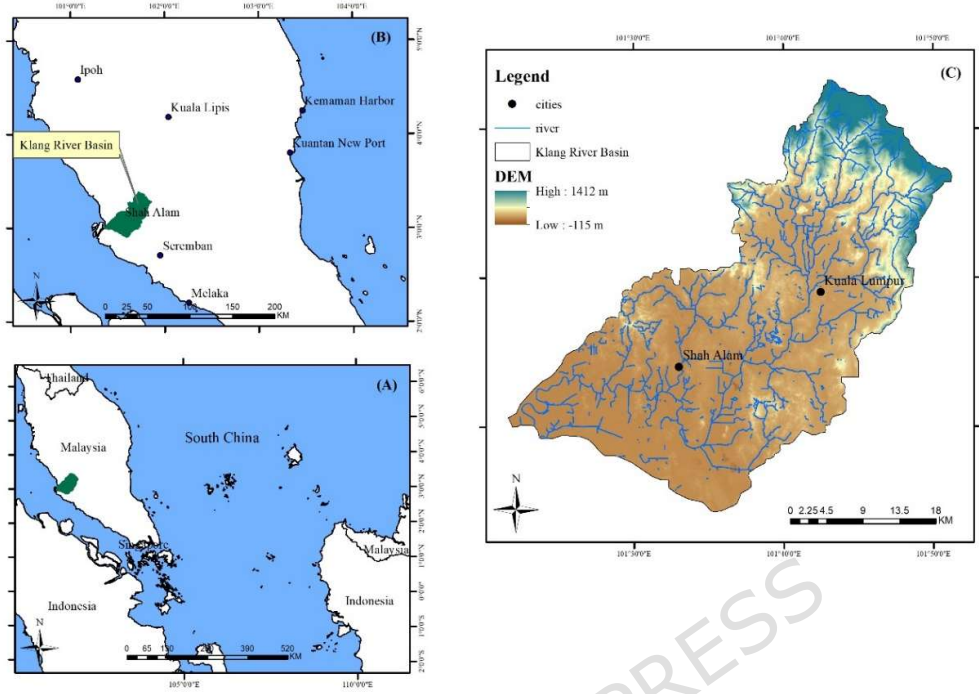


Figure 2. Study area: Klang River Basin, Peninsular Malaysia

The map was prepared by the authors using ArcGIS Pro software, version 3.5.2 (Esri, Redlands, CA, USA; <https://www.arcgis.com>), under a Sunway University institutional license.

2.2 Data Sources and Structure

This study utilizes multi-year geospatial datasets to support spatio-temporal forecasting of environmental variables over the basin. The primary dataset is annual multi-band satellite imagery (5 bands), which serves as the main prediction target. Two additional types of covariates were incorporated:

- Dynamic auxiliary variable(s): annually varying remote-sensing products (e.g., NDVI) aligned to the Landsat grid.
- Static auxiliary variables: DEM, slope, and distance-to-feature rasters (e.g., roads and rivers), treated as time-invariant spatial constraints.

All datasets span 2000–2024, with 2000–2023 used for model development and 2024 reserved exclusively for a fully unseen test.

2.3 Data Preprocessing

All geospatial inputs were resampled and aligned to a common longitude–latitude grid with

consistent spatial resolution and extent forming a regular grid of size $H \times W$, ensuring one-to-one correspondence across satellite bands, indices, and auxiliary variables. All datasets were organized into annual steps. Temporal consistency was enforced such that each grid cell maintained a continuous yearly record. Years with incomplete coverage were excluded to form a balanced spatio-temporal tensor suitable for sequence learning.

To support neighborhood-based modeling, particularly in the NA-LSTM framework, border cells were explicitly handled to ensure that every cell possessed a complete neighborhood context. The study area was treated as a closed square grid to avoid edge bias and ensures consistent spatial dependency modeling across the domain.

All undefined, missing, or invalid pixel values arising from sensor gaps, cloud contamination, or preprocessing artifacts were replaced with zeros prior to normalization. This strategy helps to avoid propagation of NaN and to keep the numbers stable during convolutional and graph-based message passing. After checking that it was not a significant proportion of the data and that no systematic spatial bias is introduced by the absence, zero-filling was selected as a neutral placeholder.

Each annual snapshot was structured into multi-channel tensors, where channels correspond to spectral bands and auxiliary variables. For NA-LSTM models, spatial neighborhoods were later unfolded into feature vectors, while for graph-based models, the same gridded data were converted into node-feature matrices.

Importantly, all models shared the same prepared input data, differing only in how spatial and temporal relationships were encoded. This common data preparation pipeline ensures that performance differences arise from model architecture rather than data inconsistencies.

Static variables (DEM, slope, and distance-to-road/river rasters) were evaluated to avoid over-parameterization and reduce weak predictors. The association between each factor and the target spatial pattern was quantified using Cramer's V in IDRISI TerrSet. Following [Eastman \(2009\)](#), predictors with Cramer's $V \geq 0.40$ were considered strong and retained, predictors with $V < 0.15$ were excluded, and intermediate values were evaluated based on interpretability and spatial relevance. Only retained variables were included in the “with auxiliary” configurations.

2.4 Forecasting Setup and Experimental Design

For supervised forecasting, a one-step-ahead temporal pairing strategy was adopted uniformly across all models. Input–output pairs were constructed as:

$$\mathbf{X}_t \rightarrow \mathbf{Y}_{t+1}$$

with samples from 2000→2001 through 2022→2023 used for training, and the final 2023→2024

transition reserved exclusively for testing.

This setting simulates an operational scenario where the model must forecast a future year without exposure to its conditions during training.

Let T denote the number of temporal layers used in the architecture:

- $T = 1$: uses 1 lstm layer or 1 temporal attention layer
- $T = 2$: uses 2 lstm layers or 2 temporal attention layers

All models were trained using the same preprocessing pipeline, normalization rules, and train/test split; therefore, performance differences primarily reflect architectural choices. Three model families were evaluated:

1. LSTM-only with CA temporal baselines
 - LSTM without auxiliary variables (spectral bands only)
 - LSTM with auxiliary variables (spectral bands + NDVI + static factors)
2. GAT–LSTM hybrid (spatial attention + temporal recurrence)
 - Spatial encoding via Graph Attention (S-GAT = 1 or 2), followed by LSTM temporal modeling ($T = 1$ or 2 steps).
3. GAT + Temporal Attention (fully attention-based temporal modeling)
 - Spatial encoding via GAT (S-GAT = 1 or 2), temporal modeling via multi-head self-attention ($T = 1$), tested with and without an MLP/FFN block.

This design isolates three key questions:

- (i) how much auxiliary data improves long-term forecasting,
- (ii) whether spatial attention improves performance beyond temporal-only learning, and
- (iii) whether temporal attention requires an MLP/FFN block to be effective in capturing the complex non-linearity of the temporal pattern in the data.

For clarity, the short model identifiers used throughout the figures and results (model_1–model_5) and their corresponding full architectural configurations are summarized in Table 1.

Table 1. Mapping between short model identifiers and full model configurations (architecture type, auxiliary inputs, spatial GAT depth S-GAT, and temporal window length T).

Short Name	Full Model Name
Model_1	GAT–Temporal Attention (MLP) S-GAT = 2, $T = 1$
Model_2	GAT–Temporal Attention (MLP) S-GAT = 1, $T = 1$
Model_3	NA-LSTM (with auxiliary inputs) S-GAT = –, $T = 1$
Model_4	NA-LSTM (without auxiliary inputs) S-GAT = –, $T = 1$
Model_5	GAT–Temporal Attention (MLP, with auxiliary inputs) S-GAT = 1, $T = 1$

All models are trained on the same partition of the input data, the same normalization method, the same target variables, the same loss functions and the same metrics for assessing their performance to make the models fairly comparable. The difference between models is their

ability and training style. This does not imply the comparison should be interpreted as a general comparison to all LSTM/GAT/attention models, but as a general comparison among the implementations and hyperparameter settings used for the specific model. Another great thing about the NA-LSTM model is that it does not need to be hyperparameter-tuned and is a lot simpler compared to hybrid GAT models.

Table 2. Hyperparameter settings are used for each model family.

Model	Hyperparameter	Value	What it controls
Models 1,2,5	hidden_channels	64	Shared hidden width for GAT + temporal-attention blocks
	num_gat_layers	2/1	Depth of spatial GAT stack
	num_temporal_layers	1	Depth of temporal self-attention stack
	gat_heads	1	Number of attention heads in each GAT layer
	temporal_heads	8	Number of heads in temporal self-attention
	gat_concat	True	Whether GAT multi-head outputs are concatenated
	dropout	0.1	Dropout strength for regularization
	learning_rate	0.01	Step size for weight updates
	num_epochs	15	Number of full training passes
	batch_size	1	Samples processed per optimization step
Models 3,4	lstm_units	32	The memory/capacity of the LSTM layer; more units allow learning more complex temporal-feature patterns.
	lstm_input_timesteps	1	How many time steps are fed per sample; with 1, the model sees only a single-step sequence.
	dense_hidden_units	16	Width of the intermediate fully connected layer; controls post-LSTM feature-combination capacity.
	dense_hidden_activation	relu	Nonlinearity in the hidden Dense layer; controls how complex/nonlinear the hidden mapping can be.
	output_units	5	Number of target outputs predicted per sample (here, 5 output channels/variables).
	output_activation	sigmoid	Output value range/behavior; constrains each

			output toward (0, 1).
	optimizer	adam	How weights are updated during training; controls update rule, adaptive learning behavior, and convergence dynamics.
	loss	huber	The objective minimized during training; balances MSE-like behavior with robustness to outliers.
	epochs	15	Number of epochs done for training
	batch_size	1024	Number of full passes over training data; controls total training duration.

Table 2 provides the parameters set for training the five models, as can be seen the number of epochs applied in the NA_LSTM is one-third of the hybrid GAT models.

2.4.1 Temporal Robustness Validation

Apart from the last holdout transition for 2023→2024, a second method called “rolling origin” was adopted for the temporal validation, in order to find the stability of the rankings. Here the models are evaluated with the data of the next year, after training the models on the data up to year t . The tested transitions include 2018→2019, 2019→2020, 2020→2021, and 2023→2024. The last transition, 2023→2024, continued being the focus, while the rest were used in measuring stability of conclusions made. Each of the rolling cross-validation has a R^2 computed.

2.4.2 Uncertainty quantification through repeated training

The models were trained with five separate random seeds to estimate prediction uncertainty and training stability. The entire training process was repeated for each seed with the same train–test split and hyperparameter settings. This last reported performance was mentioned as the mean and standard deviation of R^2 for test case 2023→2024 across 5 runs. This way one can get an empirical estimation of the uncertainty of the model related to random initialization and stochastic optimization.

2.4.3 Robustness assessment under noise and multi-step input forecasting

To assess the robustness of the system under degraded inputs, the normalized 2023 input was purposely perturbed before the 2024 prediction was made. The normalized values of the channels of the input were perturbed with the addition of gaussian noise of standard deviation 0.01, 0.03, and 0.05 and then the values were clipped between 0 and 1. Additionally, a 2 step input frame was provided to every year, for the prediction of the next year evaluated for the test hold out of

2023→2024. The performance of the model was tested against 2024 (no perturbation) data for each perturbation scenario to assess the model performance with R^2 .

2.5 Feature Engineering

2.5.1 Spatial Neighborhood Encoding (Neighborhood Augmentation Formulation)

If the aim is to explicitly capture local spatial interactions, a cellular automaton-inspired neighborhood encoding can be applied using a fixed 3×3 sliding window over the longitude–latitude grid. For a grid cell located at spatial index (i, j) , the neighborhood set $N_{i,j}$ is defined as:

$$N_{i,j} = \{(i + \Delta i, j + \Delta j) | \Delta i, \Delta j \in \{-1, 0, 1\}\} \quad (1)$$

Let $X_t^{(k)}(i, j)$ denote the value at year t and depth layer $k \in \{1, \dots, 5\}$. The spatially expanded feature vector for cell (i, j) at time t is constructed as:

$$x_t^{(k)}(i, j) = \left[X_t^{(k)}(p, q) \right]_{(p,q) \in N_{i,j}} \quad (2)$$

Stacking all depth layers yields the full spatial feature vector $x_t(i, j) \in \mathbb{R}^{5 \times 9}$. This formulation preserves local spatial coherence while remaining computationally tractable and aligns with cellular automata principles, where state evolution depends on neighboring cells.

2.5.2 Auxiliary Feature Integration

In addition to spatial features, a set of auxiliary variables was incorporated to enhance predictive performance. These auxiliary features consist of:

- Dynamic auxiliary features $a_t^{(d)}$, which vary annually (e.g., climate-driven or temporal covariates).
- Static auxiliary features $a^{(s)}$, which remain constant across time (e.g., geographic or structural attributes).

All auxiliary features were normalized to $[0, 1]$ and concatenated with the spatial feature vector:

$$\tilde{x}_t(i, j) = \left[x_t(i, j), a_t^{(d)}(i, j), a^{(s)}(i, j) \right] \quad (3)$$

In the end we will have two LSTM_CA model, one with the augmented vector as input and the other using only the pure features as input (without auxiliary data).

2.5.3 Temporal Pairing Strategy

Supervised learning samples were constructed using a one-step-ahead temporal mapping:

$$\tilde{x}_t \rightarrow y_{t+1} \quad (4)$$

where $t \in [2000, 2022]$ for training, and $\mathbf{y}_{t+1}$ corresponds to the five-layer geospatial state at year $t + 1$. The final pair $2023 \rightarrow 2024$ was reserved exclusively for model testing.

2.5.4 Data Normalization

Each input channel was normalized independently to ensure numerical stability. Given an original data value x from channel c , normalization to the range $[0, 1]$ was performed as:

$$x_{norm}^{(c)} = \frac{x^{(c)} - \min(x^{(c)})}{\max(x^{(c)}) - \min(x^{(c)})} \quad (5)$$

where $\min(\cdot)$ and $\max(\cdot)$ denote the minimum and maximum values of channel c computed across all spatial locations and all years in the training set. Missing values were replaced with zeros prior to normalization.

2.6 LSTM-Based Forecasting Model

2.6.1 LSTM Formulation

Temporal dependencies were learned using a Long Short-Term Memory (LSTM) layer. For each time step t , the updates are defined as:

$$\mathbf{i}_t = \sigma(\mathbf{W}_i \mathbf{x}_t + \mathbf{U}_i \mathbf{h}_{t-1} + \mathbf{b}_i) \quad (6)$$

$$\mathbf{f}_t = \sigma(\mathbf{W}_f \mathbf{x}_t + \mathbf{U}_f \mathbf{h}_{t-1} + \mathbf{b}_f) \quad (7)$$

$$\tilde{\mathbf{c}}_t = \tanh(\mathbf{W}_c \mathbf{x}_t + \mathbf{U}_c \mathbf{h}_{t-1} + \mathbf{b}_c) \quad (8)$$

$$\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tilde{\mathbf{c}}_t \quad (9)$$

$$\mathbf{o}_t = \sigma(\mathbf{W}_o \mathbf{x}_t + \mathbf{U}_o \mathbf{h}_{t-1} + \mathbf{b}_o) \quad (10)$$

$$\mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{c}_t) \quad (11)$$

where $\sigma(\cdot)$ is the sigmoid function, $\odot$ is element-wise multiplication, and $\mathbf{W}, \mathbf{U}, \mathbf{b}$ are trainable parameters.

2.6.2 Output Mapping and Loss Function

The final hidden state is passed through a ReLU layer followed by a sigmoid output layer:

$$\hat{\mathbf{y}}_{t+1} = \sigma(\mathbf{W}_o(\text{ReLU}(\mathbf{W}_h \mathbf{h}_t + \mathbf{b}_h)) + \mathbf{b}_o) \quad (12)$$

Training minimizes the Huber Loss: $\mathcal{L}_\delta(y, \hat{y}) = \begin{cases} \frac{1}{2}(y - \hat{y})^2 & |y - \hat{y}| \leq \delta \\ \delta|y - \hat{y}| - \frac{1}{2}\delta^2 & \text{otherwise} \end{cases} \quad (13)$

2.6.3 Post-Processing and Denormalization

The LSTM produces normalized predictions $\hat{y} \in \mathbb{R}^5$ for each location. These were reshaped to the original grid:

$$\hat{Y}_{t+1}(i, j, k) = \text{reshape}(\hat{y}_{t+1}) \quad (14)$$

To restore physical units, inverse normalization was applied:

$$\hat{X}_{t+1}^{(c)} = \hat{X}_{norm}^{(c)} \cdot (\max(X^{(c)}) - \min(X^{(c)})) + \min(X^{(c)}) \quad (15)$$

2.7 Final Forecast and Performance Evaluation

2.7.1 Final Forecast Generation

Applying the model f_θ to the 2023 inputs produces the 2024 spatial forecast:

$$\hat{x}_{2024} = f_\theta(\tilde{x}_{2023}) \quad (16)$$

Performance was quantified using Mean Absolute Error (MAE) across all spatial locations and depth layers.

2.7.2 Performance Evaluation and Vectorization Strategy

To ensure a consistent and comprehensive evaluation of model performance, all predicted and reference geospatial outputs for the year 2024 were processed in a vectorized form prior to metric computation. Specifically, for each spectral band and derived index, the two-dimensional raster output ($H \times W$) was flattened into a one-dimensional vector. The flattened vectors corresponding to all bands were then concatenated to form a single unified prediction vector. The same procedure was applied to the ground-truth dataset (Landsat2024.nc), resulting in a corresponding reference vector.

Pixels with missing or undefined values (e.g., NaN or masked values) were removed from both the prediction and reference vectors to ensure equal length and one-to-one correspondence between samples. Performance metrics, including R^2 , RMSE, MAE, MAPE, and correlation, were then computed between these two aligned vectors. This strategy enables a global, pixel-wise evaluation across all bands and spatial locations, providing a robust and unbiased assessment of overall predictive performance.

2.8 Graph-Based Spatio- Temporal Models

2.8.1 Data Pipeline

All graph-based temporal models in this work (including LSTM-GAT and GAT-Temporal Attention) use an identical data ingestion and preprocessing pipeline to ensure fair comparison. This pipeline converts multi-band satellite rasters into a standardized graph-temporal tensor representation.

2.8.1.1 Data Ingestion and Cleaning

The input dataset consists of yearly NetCDF files spanning 2000–2024:

- **Landsat:** Annual geospatial imagery with 5 spectral bands (the prediction target).
- **Dynamic Auxiliary (NDVI):** Annual NDVI rasters aligned to the Landsat grid.
- **Static Auxiliary:** DEM, road distance, and slope rasters (time-invariant).

All undefined or missing grid values are replaced with zeros before normalization. This prevents NaN propagation through graph operations.

2.8.1.2 Channel-wise Normalization

All channels are normalized using channel-wise min–max scaling

$$x_{norm}^{(c)} = \frac{x^{(c)} - \min(x^{(c)})}{\max(x^{(c)}) - \min(x^{(c)})} \quad (17)$$

where $\min(\cdot)$ and $\max(\cdot)$ are computed per channel (c) over the training data distribution.

2.8.1.3 Temporal Pairing and Spatial-to-Graph Conversion

A one-step-ahead forecasting formulation is used. Training pairs span (2000 → 2001) to (2022 → 2023), with (2023 → 2024) held out for testing. Each raster grid is converted into a graph $G = (V, E)$:

- **Nodes (V):** Each pixel becomes a node ($\|V\| = H \times W$).
- **Edges (E):** Connect spatial neighbors using 4-connectivity.
- Graph size is therefore defined by $H \times W$ nodes and grid-local edges based on 4-connectivity. In memory, the graph is represented using sparse COO edge indices, and training is performed with batch size 1 to reduce CPU memory usage.
- **Node features:** For each year t , node i receives a feature vector $x_{i,t} \in \mathbb{R}^F$, where $F=5$ (Landsat) or larger (with auxiliary features).

2.8.2 Graph-Based Spatio-Temporal Forecasting Architectures

To benchmark graph neural approaches, two architectures were implemented using the same graph-structured Landsat input representation.

2.8.2.1 GAT–LSTM: Integration of Spatial Attention with Temporal Recurrence

The GAT–LSTM architecture performs spatial feature extraction first using Graph Attention Networks (GAT), followed by temporal modeling using an LSTM.

Spatial module (GAT) For each time step (t), node features $h_{i,t}^{(0)}$ are updated by attention-weighted aggregation from neighbouring nodes $j \in N(i)$. The attention coefficient $e_{ij,t}$ and normalized weight $\alpha_{ij,t}$ are computed as:

$$e_{ij,t} = \text{LeakyReLU} \left(a^\top [Wh_{i,t}^{(0)} \parallel Wh_{j,t}^{(0)}] \right) \quad (18)$$

$$\alpha_{ij,t} = \frac{\exp(e_{ij,t})}{\sum_{k \in \mathcal{N}(i)} \exp(e_{ik,t})} \quad (19)$$

The spatially aggregated node embedding is: $h_{i,t}^{(GAT)} = \sigma \left(\sum_{j \in \mathcal{N}(i)} \alpha_{ij,t} Wh_{j,t}^{(0)} \right)$ (20)

where $\parallel$ denotes concatenation, $\sigma(\cdot)$ is a non-linear activation, and W, α are learned parameters.

Temporal module (LSTM) For each node i , the sequence $\{h_{i,t}^{(GAT)}\}$ across years is passed into an LSTM:

$$h_{i,t}^{(LSTM)} = \text{LSTM}(h_{i,t}^{(GAT)}, h_{i,t-1}^{(LSTM)}) \quad (21)$$

The final state is mapped via a linear layer to predict the next-year Landsat bands:

$$\hat{y}_{i,t+1} = W_o h_{i,t}^{(LSTM)} + b_o \quad (22)$$

2.8.2.2 GAT + Temporal Attention: Integration of Spatial and Temporal Self-Attention

This architecture is fully attention-based, replacing recurrence with multi-head self-attention.

Spatial module (GAT) Identical to the GAT stage described in Model 1, producing node embeddings $h_{i,t}^{(GAT)}$ for each year.

Temporal module (Multi-head self-attention) For each node (i), the temporal sequence $H_i = [h_{i,1}^{(GAT)}, \dots, h_{i,T}^{(GAT)}]$ is processed using scaled dot-product attention. Queries, keys, and values are

$$Q = H_i W_Q, \quad K = H_i W_K, \quad V = H_i W_V \quad (23)$$

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^\top}{\sqrt{d_k}} \right) V \quad (24)$$

For multi-head attention (MHA), outputs from H heads are concatenated and projected.

Optionally, a feed-forward block (FFN) is applied:

$$\text{FFN}(x) = W_2 \phi(W_1 x + b_1) + b_2 \quad (25)$$

The final representation z_i is projected to the five-band output:

$$\hat{y}_{i,t+1} = W_o z_i + b_o \quad (26)$$

2.8.2.3 Depth Configurations

To manage complexity, depth combinations were constrained:

GAT layers	Temporal Layers
2	1
1	2

2.8.3 Summary of Model Differences

- **GAT-LSTM:** spatial attention + sequential temporal memory.

- **GAT + Temporal Attention:** spatial attention + parallel, non-recurrent Transformer-style attention (evaluated with and without MLP).

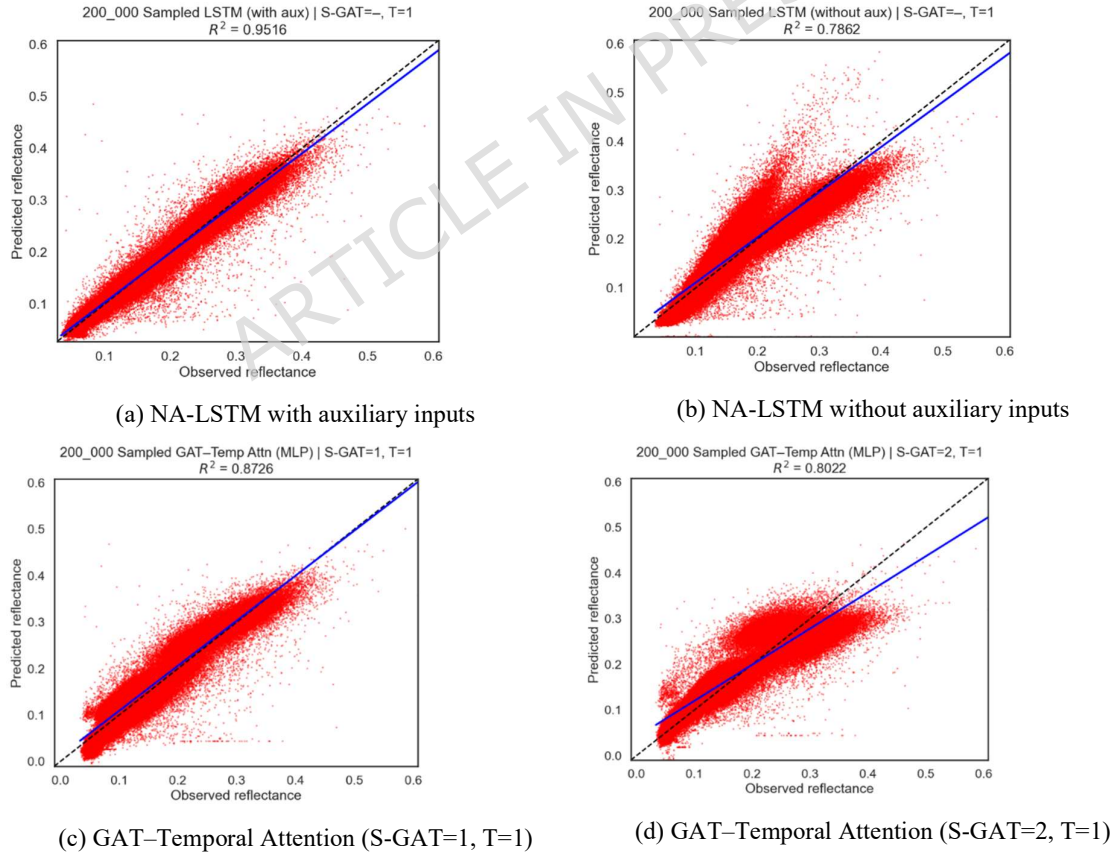
Both models operate on the same graph representation, enabling direct performance comparison.

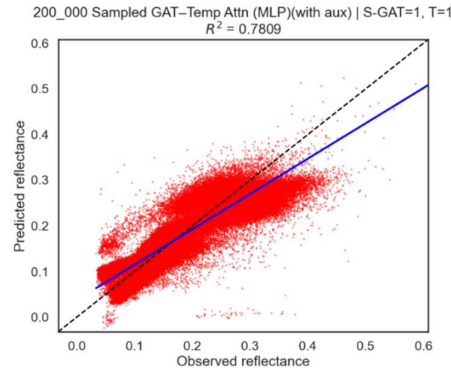
4. Case Study

4.1 One-Year-Ahead Forecasting Performance (2023 → 2024)

All models were evaluated on the held-out transition from 2023 to 2024, which serves as a strict one-year-ahead unseen-year test. Identical data preprocessing, normalization, and training protocols were applied to all architectures to ensure a fair comparison. Performance was assessed using multiple error and agreement metrics computed separately for each index. Figure 4 presents scatter plots comparing predicted and observed values for representative models, illustrating the agreement, dispersion, and bias characteristics under the one-year-ahead forecasting setting.

The LSTM model with auxiliary inputs shows the tightest clustering around the 1:1 line, whereas attention-based and hybrid models exhibit larger spread and more pronounced systematic deviations, consistent with their lower quantitative scores.





(e) GAT-Temporal Attention with auxiliary inputs

Figure 4: Scatter plots of predicted vs. observed values for representative models

Clear performance differences were observed across model families. The LSTM with auxiliary inputs ($T = 1$) achieved the highest overall accuracy, with consistently strong performance across most indices and an average R^2 of approximately 0.95. The GAT-Temporal Attention model with an MLP block ($S\text{-GAT} = 1, T = 1$) ranked second, achieving an average R^2 of approximately 0.87. In contrast, the GAT-Temporal Attention model without the MLP and the LSTM-GAT hybrid models exhibited substantially weaker performance, with R^2 values around 0.34 or lower for several indices. These results indicate a strong dependence of forecasting skill on both architectural design and input configuration.

4.2 Model-Level Performance Comparison

Aggregating performance across indices reveals a clear ranking among the tested approaches. The LSTM with auxiliary inputs consistently outperformed all other models in terms of RMSE, MAE, correlation, and R^2 . The LSTM without auxiliary input achieved the second-best overall performance, but with reduced explanatory power.

The GAT-Temporal Attention model with MLP showed competitive performance, particularly for vegetation- and water-related indices, but remained inferior to the LSTM with auxiliary inputs. For the single 2024 unseen-year test, the GAT-Temporal Attention model with $S\text{-GAT} = 1$ achieved stronger performance than the $S\text{-GAT} = 2$ configuration. However, under rolling-origin validation across 2019, 2020 and 2021 test years, the $S\text{-GAT} = 2$ GAT-Temporal Attention model achieved the second-highest average R^2 , indicating better temporal robustness across multiple test years.

The LSTM-GAT hybrid models showed unstable behavior across configurations, with several indices exhibiting negative R^2 values, indicating performance worse than a mean-value baseline. Similarly, the GAT-Temporal Attention model without MLP consistently underperformed relative to its MLP-equipped counterpart. [Figures 5–8](#) summarize the model-level performance

across all indices using R^2 , RMSE, MAE, and correlation, respectively. The LSTM with auxiliary inputs consistently achieves the best scores across all metrics, followed by the LSTM without auxiliary inputs and the GAT–Temporal Attention model with an MLP block. Hybrid LSTM–GAT and attention-only models without an MLP show markedly inferior and less stable performance.

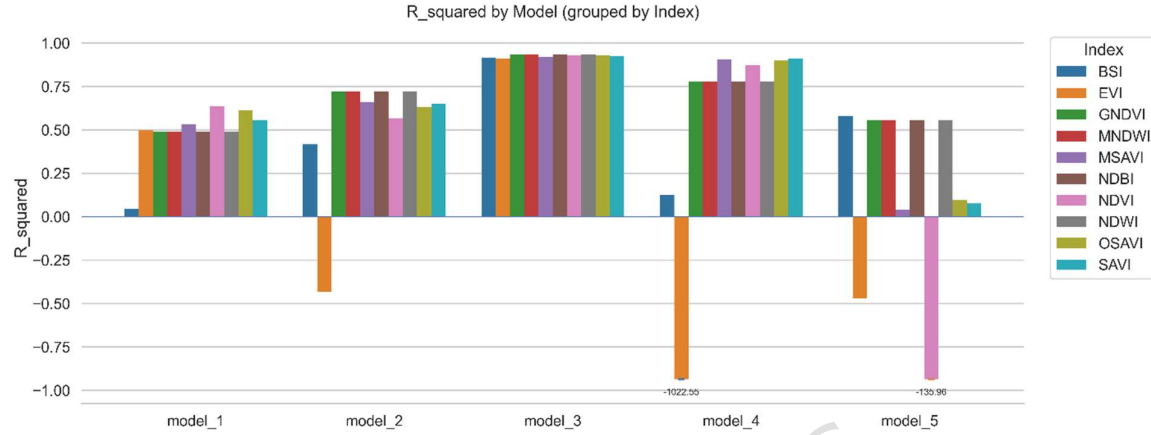


Figure 5. Comparison of coefficient of determination (R^2) across models and spectral indices for the one-year-ahead (2023→2024) forecasting task. Bars show index-wise R^2 values for each model, highlighting differences in predictive skill and stability.

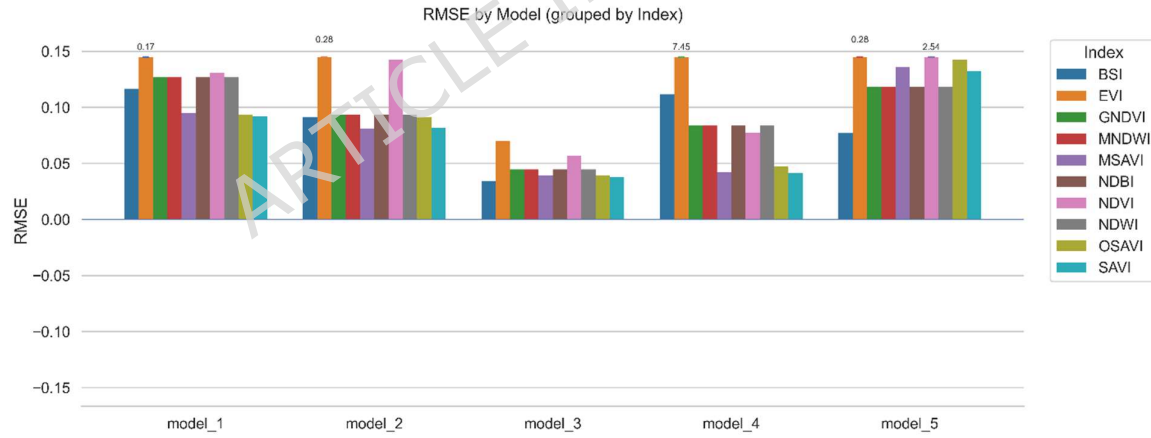


Figure 6. Comparison of root mean square error (RMSE) across models and spectral indices for the one-year-ahead (2023→2024) forecasting task. Lower values indicate better predictive accuracy, and the grouped bars illustrate index-wise error variations across model architectures.

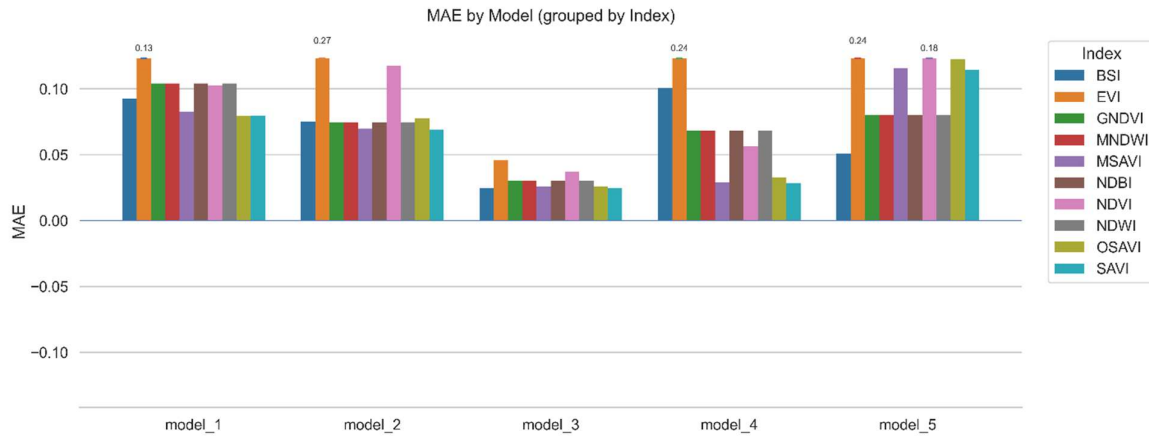


Figure 7. Comparison of mean absolute error (MAE) across models and spectral indices for the one-year-ahead (2023→2024) forecasting task. Lower values indicate better predictive accuracy, and the grouped bars illustrate index-wise error variations across model architectures.



Figure 8. Comparison of Pearson correlation coefficients between predicted and observed values across models and spectral indices for the one-year-ahead (2023→2024) forecasting task. Higher values indicate stronger linear agreement, whereas negative values indicate inverse or failed predictive relationships. As attention-based models are more sensitive to depth, learning rate, regularization, and graph construction, their weaker performance should be interpreted in the context of the selected hyperparameter space rather than as a universal limitation of attention mechanisms.

The negative R^2 values for some model–index combinations should be considered as a sign of weak predictive generalization, and not as an error in the calculations. When the prediction error is greater than the error from the average value of the observed test values, the value of R^2 is negative. For this study, the temporal variation or value range was limited for some of the indicators during the unseen test period. In such a case, moderate prediction errors could result in R^2 of a very negative value. Furthermore, various auxiliary indices don't react in the same way to land surface changes over time. The temporal evolution of some indices may not be correlated

to the temporal evolution of the target year, but they may contain useful environmental information for the model. Shared spectral or environmental information may also be reflected in similar metrics across some indices, such as NDVI, SAVI, and GNDVI, are all related to the condition of the vegetation; and other indices may respond similarly to moisture, built-up intensity or surface dryness. Thus, such values were kept as diagnostic information on the instability of the model and the difficulty of predicting the index.

4.3 Temporal Robustness under Rolling-Origin Cross-Validation

Table 3. Rolling-origin cross-validation R^2 performance of the five model configurations across test years 2024, 2021, 2020, and 2019. For each test year, models were trained using all available years up to the preceding year and tested on the following year.

Model	Test 2024 R^2	Test 2021 R^2	Test 2020 R^2	Test 2019 R^2	Average R^2
Model 3	0.948 $\pm$ 0.021	0.906	0.915	0.882	0.913
Model 1	0.792 $\pm$ 0.033	0.932	0.904	0.894	0.880
Model 5	0.754 $\pm$ 0.038	0.873	0.874	0.895	0.856
Model 2	0.842 $\pm$ 0.048	0.835	0.741	0.806	0.814
Model 4	0.777 $\pm$ 0.028	0.712	0.701	0.761	0.740

Table 3 confirms that the main performance ranking is temporally stable across multiple rolling-origin test years. The NA-LSTM model with auxiliary inputs achieved the highest average R^2 of 0.913 despite having substantially fewer trainable parameters than the graph-attention models. This suggests that the 3×3 neighborhood augmentation and auxiliary environmental covariates provide a strong inductive bias for the annual geospatial forecasting task. The GAT–Temporal Attention model with two GAT layers achieved the second-highest average R^2 of 0.880, outperforming the corresponding one-layer GAT configuration without auxiliary inputs. This indicates that, although the 2024 holdout result favored the one-layer GAT model, the two-layer GAT configuration was more robust when evaluated across multiple rolling test years. This difference suggests that the S-GAT = 1 configuration may be better fitted to the specific spatial–temporal conditions of the 2023→2024 transition, while S-GAT = 2 provides a more stable representation when performance is averaged across several temporal splits. Therefore, the apparent contradiction reflects a difference between single-year peak performance and multi-year temporal robustness, rather than an inconsistency in the results. The comparison also shows that auxiliary inputs improved the average performance of both matched model families: NA-LSTM increased from 0.740 to 0.913, and the one-layer GAT–Temporal Attention model increased from

0.814 to 0.856. However, the lower 2024 R^2 of the auxiliary GAT model suggests that the benefit of auxiliary variables may vary by test year and architecture.

For the year 2024 the repeated-seed experiment showed that the LSTM model with auxiliary inputs maintained the highest mean performance and the lowest performance variability, indicating that its superior performance was not caused by a single favorable initialization.

4.4 Index-Wise Performance Characteristics

Across models, systematic differences were observed between index groups. Vegetation- and water-related indices — including the Normalized Difference Vegetation Index (NDVI), Green Normalized Difference Vegetation Index (GNDVI), Soil-Adjusted Vegetation Index (SAVI), Modified Soil-Adjusted Vegetation Index (MSAVI), Optimized Soil-Adjusted Vegetation Index (OSAVI), Normalized Difference Water Index (NDWI), Modified NDWI (MNDWI), and Normalized Difference Built-up Index (NDBI) — were generally predicted with higher accuracy, particularly by the LSTM-based models, as reflected by high correlation and high R^2 values. As shown in Figures 5–8, these indices consistently achieve higher R^2 and lower errors across most models, indicating greater temporal predictability and stability.

In contrast, EVI and BSI exhibited larger errors and greater variability across models. For several architectures, these indices showed reduced correlation and, in some cases, negative R^2 values, indicating lower predictability under the one-year-ahead forecasting setting and higher sensitivity to noise and scaling effects.

4.5 Ablation and Architectural Comparisons

The ablation effects discussed above are quantitatively summarized in Figures 5–8, which reveal consistent performance differences attributable to the inclusion of auxiliary inputs, the depth of spatial attention, and the presence of the MLP/FFN block.

Effect of auxiliary inputs. Including auxiliary variables led to a substantial improvement in performance. The LSTM with auxiliary inputs achieved an average R^2 of approximately 0.95, whereas the same architecture without auxiliary inputs achieved an average R^2 of approximately 0.79, highlighting the importance of external driving factors for long-term forecasting.

Effect of spatial attention depth. For the 2023→2024 holdout test, S-GAT = 1 outperformed S-GAT = 2; however, the rolling-origin validation showed that S-GAT = 2 achieved better average temporal robustness across multiple test years. This suggests that shallower spatial

attention may be more effective for the specific 2024 transition, whereas the deeper GAT configuration may capture more stable spatial dependencies across different temporal splits.

Effect of the MLP/FFN block in temporal attention. Removing the MLP/FFN block from the temporal attention architecture resulted in a pronounced performance drop, with the average R^2 decreasing to approximately 0.34, indicating that the feed-forward component is critical for effective temporal representation learning.

Hybrid LSTM–GAT architectures. The LSTM–GAT models exhibited inconsistent results across configurations. One configuration (S-GAT = 2, T = 1) achieved moderate performance ($R^2 \approx 0.61$), whereas another (S-GAT = 1, T = 2) showed substantially lower performance ($R^2 \approx 0.24$), indicating sensitivity to architectural design choices and training stability.

4.6 Computational Complexity and Scalability

An operational geospatial forecasting scenario is one where the number of pixels may be quite large when expanding the spatial resolution or the extent of the study area, so computational scalability is important. The LSTM–CA model adopts a fixed 3×3 neighborhood representation; thus its input size increases linearly with the number of pixels and channels. By comparison, graph-based models involve building and fitting graph representations of the nodes and edges. The number of nodes is $H \times W$ (1825, 1867) and the number of edges is about $4HW$ (excluding boundary effects) for a regular grid with four-neighbor connectivity. GAT-based models thus add extra memory and computational cost in computing attention weights for graph edges. There are also attention operations across temporal steps which are a requirement for temporal attention models, but which are not so expensive in the current study as the length of the annual sequence is relatively short. Table 4: model complexity of the different model architectures

Model	Spatial representation	Approximate scaling	Main cost
NA-LSTM	Local 3x3 neighbourhood per pixel	$O(HW * 9C)$ (neighbourhood build) + $O(HW * T * (h * (9C) + h^2))$ (LSTM)	Per-pixel sequence modelling after neighbourhood expansion
GAT-LSTM	Graph nodes/edges	$O(T * E * heads * h)$ (GAT) + $O(N * T * h^2)$ (LSTM), with $N=HW$	Edge-wise attention plus recurrent temporal updates

GAT-Temporal Attention	Graph + temporal self-attention	$O(T * E * \text{heads} * h)$ (GAT) + $O(N * T^2 * h)$ (temporal attention)	Graph attention plus quadratic-in-time self-attention
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Table 5. The number of trainable parameters of the tested model configurations.

Model configuration	Auxiliary inputs	GAT layers	Trainable parameters
GAT-Temporal Attention MLP	No	2	59,269
GAT-Temporal Attention MLP	No	1	54,981
GAT-Temporal Attention MLP	Yes	1	55,301
NA-LSTM	No	–	10,725
NA-LSTM	Yes	–	16,485

The result of the trainable parameter counts demonstrate that the NA-LSTM models are significantly simpler than the GAT-Temporal Attention models. The NA-LSTM model can be trained with just 16,485 trainable parameters, while the GAT-Temporal Attention configurations require 55,301–59,269 parameters. Even with that lower capacity, it was able to get the highest average rolling-origin R^2 . This is consistent with the understanding that for this dataset, the ability to make a spatial inductive bias and having additional environmental information is more important than increasing the capacity of the graph-attention model.

All experiments were performed on CPUs instead of GPUs because the graph-temporal data representation and full-raster processing needs a lot of system memory, which was around 35–40 GB at the maximum of the memory used in the model runs. In this implementation, the numbers of large floating-point tensors in the experiments were sufficiently large, and the CPU execution also made it possible to obtain stable numerical executions with enough capacity of memory. This means that when selecting a CPU, it should be considered as a constraint on hardware-memory and implementation, not as a reason to believe that CPUs are generally faster or more efficient than GPUs for floating point based deep learning workloads.

4.7 Uncertainty and Robustness Analysis and multi-step horizon

Table 6. Performance results of the 2023→2024 holdout test for the five model configurations under two-step input forecasting and Gaussian input-noise perturbations presented as R^2 values. The first column of R^2 in the table is for the prediction for 2023 from 2024, based on a two-step prediction frame, and the second column is the prediction for 2024 without adding the noise. The

rest of the columns show the performance when (0, 0.01, 0.03, 0.05) zero-mean Gaussian noise is added to the normalized input data.

Table 6. Model Performance and Noise Robustness in the 2023–2024 Holdout Test

Model	Two-step input R^2	Original 2024 R^2 , no noise	Noise $\sigma = 0.01$	Noise $\sigma = 0.03$	Noise $\sigma = 0.05$
Model 2	0.610	0.872	0.857	0.845	0.824
Model 3	0.686	0.951	0.910	0.879	0.821
Model 1	0.678	0.802	0.776	0.772	0.764
Model 4	0.217	0.786	0.754	0.746	0.723
Model 5	0.682	0.781	0.761	0.754	0.740

Table 6 presets the five model configurations are tested under two complementary stress tests: two-step input forecasting and controlled Gaussian perturbation of the normalized 2023 input data. All models showed some degradation for use of a two-step input frame, compared to the original no-noise setting, suggesting an optimal tuning of the tested architectures for the original one-step input setting. The NA-LSTM model without auxiliary inputs experienced the greatest degradation with the R^2 changing from 0.786 to 0.217, indicating that the non-auxiliary LSTM model is very sensitive to the change in temporal input framing. Under the two-step input setting, the GAT–Temporal Attention model with auxiliary inputs had the smallest decrease (0.781 to 0.682) followed closely by the two-layer GAT–Temporal Attention model.

All models had a decrease in the R^2 with increasing noise standard deviation (0.01–0.05) under Gaussian input noise condition. The NA-LSTM model had the highest R^2 for low and moderate noise levels of 0.910 at $\sigma = 0.01$ and 0.879 at $\sigma = 0.03$. But at $\sigma = 0.05$, its performance dropped to 0.821 and is now similar to that of the one-layer GAT–Temporal Attention model without auxiliary inputs, which performed at 0.824. This means that the auxiliary NA-LSTM model is still the most accurate when the perturbation is mild but is more sensitive to high intensity noise than several GAT-based models. The absolute difference in R^2 between the two-layer GAT–Temporal Attention model and the best-performing NA-LSTM model is the lowest under noise at 0.802–0.764, indicating that GAT–Temporal Attention has the best relative noise robustness, but its absolute R^2 is still below the NA-LSTM model at low and moderate noise levels.

The robustness analysis reveals that auxiliary information contributes to a stronger predictive accuracy for the peak, but the robustness of the models depends on the type of perturbation: NA-LSTM-CA with auxiliary inputs is more robust for the original and mild-noise perturbation, while

GAT–Temporal Attention models are more robust for the strong-noise and two-step input change settings.

4.8 Metric Behavior and Outliers

MAPE values were found to be highly sensitive for indices with values near zero, leading to inflated percentage errors in some cases. In addition, negative R^2 values were observed for certain model–index combinations, indicating performance worse than a mean-value baseline. These negative R^2 values should be interpreted as evidence of weak predictive generalization rather than calculation errors. R^2 becomes negative when the prediction error is larger than the error obtained by using the mean of the observed test values as a baseline. In this study, some indices had limited temporal variation or a narrow value range during the unseen test period. Under such conditions, even moderate prediction errors can lead to large negative R^2 values.

Not all auxiliary indices respond equally to land-surface change over time. Some indices provide useful environmental information for the model, but their temporal behavior may not always be strongly linked to the observed changes in the target year. Similar metrics across some indices may also reflect shared spectral or environmental information, as indices such as NDVI, SAVI, and GNDVI are all related to vegetation conditions, while other indices may respond similarly to moisture, built-up intensity, or surface dryness. Therefore, these values were retained as diagnostic evidence of model instability and index-specific prediction difficulty.

These effects highlight the importance of interpreting MAPE and R^2 jointly with MAE, RMSE, and correlation. Figure 9 shows that MAPE exhibits large variability for indices with near-zero values, while Figures 10 and 11 show the mean and standard deviation of prediction differences, highlighting the presence of outliers and heteroscedastic error behavior for specific model–index combinations.

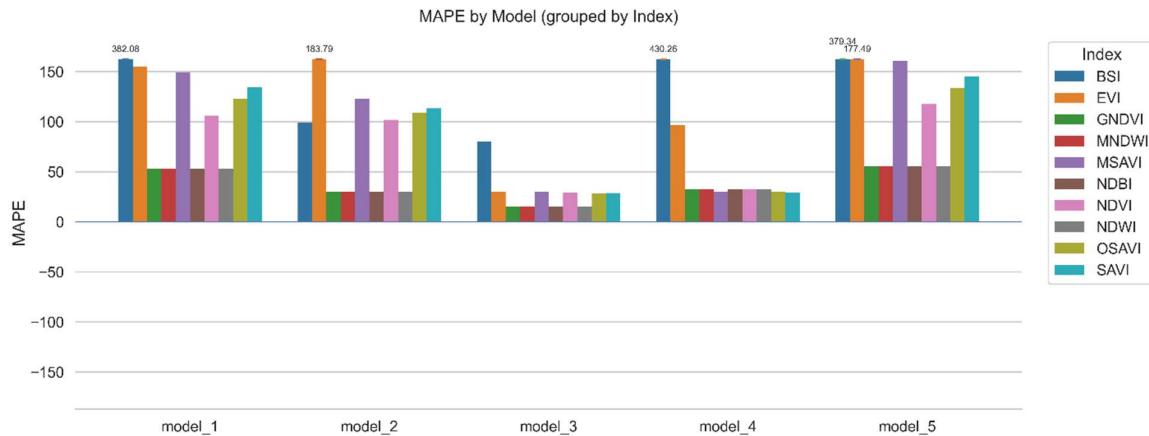


Figure 9. Comparison of mean absolute percentage error (MAPE) across models and spectral indices for the one-year-ahead (2023→2024) forecasting task. Large variations and extreme

values reflect the sensitivity of MAPE to indices with values close to zero, highlighting potential instability of percentage-based error metrics in this setting.

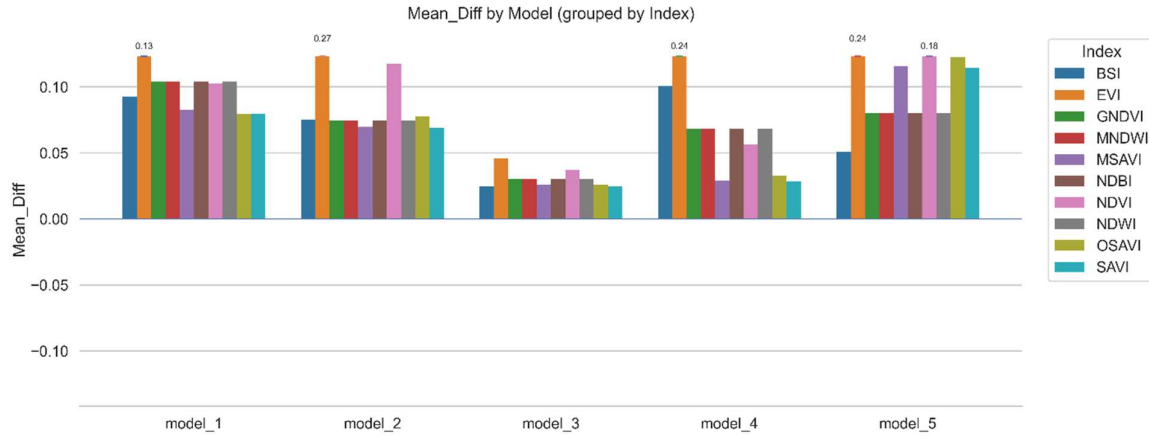


Figure 10. Comparison of mean prediction difference (bias) across models and spectral indices for the one-year-ahead (2023→2024) forecasting task. Values closer to zero indicate lower systematic bias, while larger positive or negative values indicate systematic over- or underestimation.

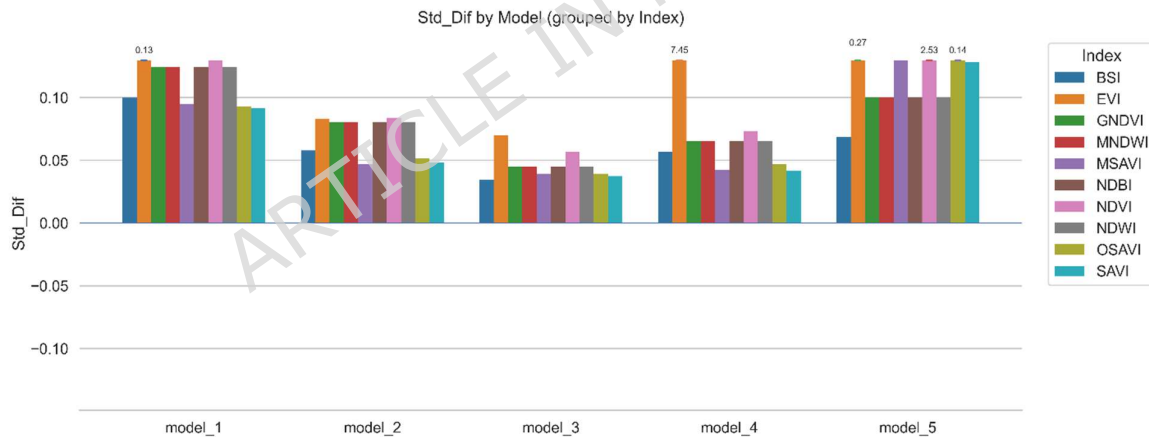


Figure 11. Comparison of the standard deviation of prediction differences across models and spectral indices for the one-year-ahead (2023→2024) forecasting task. Lower values indicate more stable and less dispersed errors, whereas higher values reflect greater variability in prediction errors across spatial locations.

5. Discussion

5.1 Temporal Context and Auxiliary Information

The better performance of the LSTM model with the addition of the auxiliary inputs confirms that the long-term geospatial dynamics are controlled by both multi-year temporal relationships

and the exogenous inputs. The performance difference between the LSTM model with and without the use of auxiliary variables suggests that using only the spectral history is not enough to capture all the inter-annual variability. Rather, static and dynamic covariates offer complementary information which increases the robustness of predictions and contributes to better generalization to future conditions that are not seen during the training process. The results provide evidence in favor of the idea that a model incorporating temporal memory and context, not just autoregressive spectral patterns, is essential for reliable long-term forecasting of the environment.

5.2 Effectiveness and Limitations of Spatial Attention

One way to understand the performance gain of the NA-LSTM model is the model's inductive bias. Specifically, the assumption embedded into the architecture of the 3×3 neighborhood-augmented input ensures that there is a strong local-spatial assumption incorporated into the model architecture. This assumption is quite similar to the local dynamics of environmental processes, where adjacent pixels have similar land cover, water and vegetation features. Unlike graph attention models, which need to learn spatial weights from a small number of samples per year, the neighborhood-augmented LSTM model gets local spatial information directly. The latter learns annual time-series dynamics, while external environmental effects are learned through auxiliary features. This generally implies that the model will have a lower learning load as it is not required to learn all the assumptions from scratch.

Spatial attention mechanisms can improve the performance of predictive modeling when suitably designed, as the results show that the GAT-Temporal Attention model with an MLP block outperforms the others. The modest improvements over the LSTM with auxiliary inputs, however, imply that, at least for the current data and forecasting duration, temporal dynamics and auxiliary drivers are more important than complex spatial message passing.

Moreover, the performance drop as the spatial GAT depth increased suggest two potential factors. The first issue is that repeated graph-attention aggregation may smooth the representation of neighboring nodes, which can diminish the local contrast between heterogeneous land-cover or environment states. The first problem is that repeated graph-attention aggregation can smooth the representation of neighboring nodes and decrease local contrast between heterogeneous land-cover or environmental states. Second, the temporal sample size is small as the data is annual, and deeper spatial-attention models will be more difficult to reliably optimize.

5.3 The significance of the MLP/FFN Block in Attention-Based Models

The significant performance gap between temporal attention models with and without the MLP/FFN block demonstrates the importance of nonlinear feature transformation in attention networks. Without this, the model performs less well in projective temporal learning in the target space and suffers from underfitting and poor generalization.

This is aligned with the standard designs of Transformer models, where the feed-forward network is built into the core of representation learning and plays a critical role for attaining the desired model expressiveness.

5.4 Hybrid LSTM–GAT Architectures and Model Stability

The instability in LSTM–GAT models may also be caused by the sequential interaction of two complex blocks, with LSTM capturing temporal transitions from these transformed representations, and graph attention capturing spatial information in a graph. The mistakes made in the encoding of spatial information can be injected into the temporal learning block, where they can be susceptible to initializations and depth of yearly satellite data with short time series.

The LSTM–GAT hybrid models performed inconsistently and often sub-optimally, suggesting that using a spatial attention extractor alongside the temporal model of LSTM layers is not sufficient to achieve good forecasting performance. These models are sensitive to the architectural depth and layer ordering, making them more challenging to optimize and harder to be trained in some situations with more instability, especially for long term extrapolation scenarios. In comparison, simpler and better-conditioned architectures, such as the LSTM with auxiliary inputs, exhibit more stable, reliable and interpretable performance in the current forecasting problem.

5.5 Index-Specific Challenges and Metric Behavior

The reduced predictive performance observed for EVI and BSI across several models suggests that these indices are more sensitive to noise, scaling effects, or complex nonlinear dynamics. This underscores the importance of index-wise evaluation rather than relying exclusively on aggregated performance metrics. Moreover, MAPE was also determined as being less reliable in the case of indices with values near zero because the absolute error is quite significant in comparison with the percentage error. In this context, therefore, it is more meaningful to examine the predictive performance of models as reflected in MAE, RMSE, correlation, and R^2 .

Lastly, some negative values for R^2 in some configurations shed light on the fact that the measures need to be calculated consistently and interpreted carefully, as R^2 values can be moderately high and show worse performance than a mean-value baseline.

This study does not conduct complete spatial-zone-wise validation by land-use class, elevation zone or ecological subregion, although it does carry out index-wise evaluation and spatial error diagnostics (prediction differences and error dispersion). Future studies should be carried out in these spatial zones to further determine the strengths and weaknesses of the models in the geographic areas.

5.6 Overfitting Risk and Limited Temporal Samples

There are relatively few time training transitions possible with consisting, as it is only observed annually from 2000-2024. This poses a risk of overfitting, especially for models that have high capacity like graph-attention and temporal-attention models. Although there will be numerous samples within the spatial dimension because of the large number of examples available that describe the interaction at a pixel level, these examples are not completely independent because of the spatial autocorrelation between them. Therefore, with respect to the model architecture and errors in the spatial domain, good performance on the left-out year should also be viewed that way. Several regularization techniques were employed during training and evaluation to avoid overfitting, dropout, early stopping, Huber loss and held-out unseen-year test. Also rolling cross-validation was conducted to test the generalization of the architecture performance.

Conclusion

This study proposed a unified benchmarking framework for one-step-ahead long-term spatio-temporal geospatial forecasting using multi-band satellite data. Using a harmonized preprocessing pipeline, consistent train-test splitting (2000–2023 for training and 2024 as a fully unseen test year), and vectorized pixel-wise evaluation, we compared neighborhood-based NA-LSTM models with and without auxiliary variables against graph-attention-based architectures, including GAT-LSTM hybrids and GAT-Temporal Attention models with ablation of the MLP/FFN block. The vectorized pixel-wise evaluation was retained as the primary benchmark because it provides a consistent basis for comparing all model configurations across the full study area and all output indices under the same testing condition.

Across all indices and metrics, the NA-LSTM model augmented with auxiliary inputs achieved the best and most stable performance (average $R^2 \approx 0.95$), demonstrating that robust temporal modeling combined with external driving factors is more critical for long-term generalization

than increasing spatial attention complexity. The results show that the effect of GAT depth depends on the evaluation setting: S-GAT = 1 performed better on the 2024 holdout test, whereas S-GAT = 2 achieved the second-highest average R^2 under rolling-origin validation.. Removing the MLP/FFN from the temporal attention architecture caused a pronounced drop in predictive skill, confirming that the feed-forward component is essential for effective representation learning in attention-based models.

Index-wise analysis further showed that vegetation- and water-related indices (e.g., NDVI, NDWI, MNDWI, SAVI) are consistently more predictable than EVI and BSI, which exhibited higher variability and sensitivity to scaling and near-zero values. In addition, MAPE was found to be unreliable for indices close to zero, highlighting the importance of interpreting percentage-based errors together with MAE, RMSE, correlation, and R^2 .

Future work should extend this benchmark in five directions. First, multi-step and long-horizon forecasting should be evaluated to quantify error accumulation over multiple future years. Second, uncertainty quantification methods such as Monte Carlo dropout, deep ensembles, or quantile prediction should be incorporated to assess forecast reliability. Third, the framework should be tested across multiple geographic regions and environmental contexts to evaluate spatial transferability. The rolling-origin validation provides evidence of temporal robustness, the analysis of this study is still limited to one geographic region. Therefore, the findings should be interpreted as region-specific rather than fully generalized across all environmental contexts. Future studies should extend the same benchmarking framework to regions with different climatic, topographic, land-use, and ecological characteristics to further test the spatial transferability of the models. Fourth, self-supervised and foundation-model approaches for Earth observation could be explored to improve representation learning when labeled temporal samples are limited. Finally, explainable AI methods, including attention-map inspection, feature attribution, and spatial error diagnostics, should be used to improve interpretability and support decision-making in environmental monitoring.

Ethics approval and consent to participate

Not applicable. This study did not involve human participants, human data, human tissue, or animals.

Consent for publication

Not applicable.

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Conflict of interest

The authors state no conflict of interest. The authors used language-editing assistance to improve clarity and readability. The scientific content remains entirely the authors' own.

Data Availability

The data used in this study are available based on request.

CREDIT AUTHORSHIP STATEMENT: Kaveh Karimadini: Data curation; Formal analysis, Software; Validation; Writing – original draft. Sharareh Pourebrahim: Funding acquisition; Methodology; Supervision; Writing – original draft;

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Table 1: Summary of Metrics

Index	RMSE	MAE	MAPE	Correlation	R_squared	Mean_Diff	Std_Dif	model_name	S-GAT	T-Layers
NDVI	0.1308	0.1026	106.1529	0.8570	0.6352	0.1026	0.1296	GAT-Temp Attn (MLP)	2	1
NDBI	0.1268	0.1042	53.0599	0.9238	0.4886	0.1042	0.1244	GAT-Temp Attn (MLP)	2	1
NDWI	0.1268	0.1042	53.0599	0.9238	0.4886	0.1042	0.1244	GAT-Temp Attn (MLP)	2	1
MNDWI	0.1268	0.1042	53.0599	0.9238	0.4886	0.1042	0.1244	GAT-Temp Attn (MLP)	2	1
EVI	0.1652	0.1294	154.8633	0.8956	0.4971	0.1294	0.1304	GAT-Temp Attn (MLP)	2	1
SAVI	0.0919	0.0798	134.5598	0.9265	0.5565	0.0798	0.0919	GAT-Temp Attn (MLP)	2	1
MSAVI	0.0949	0.0826	149.2168	0.9244	0.5324	0.0826	0.0947	GAT-Temp Attn (MLP)	2	1
BSI	0.1166	0.0928	382.0753	0.6772	0.0462	0.0928	0.0998	GAT-Temp Attn (MLP)	2	1
GNDVI	0.1268	0.1042	53.0599	0.9238	0.4886	0.1042	0.1244	GAT-Temp Attn (MLP)	2	1
OSAVI	0.0933	0.0792	122.5745	0.9183	0.6139	0.0792	0.0931	GAT-Temp Attn (MLP)	2	1
NDVI	0.1424	0.1176	101.9778	0.9234	0.5677	0.1176	0.0839	GAT-Temp Attn (MLP)	1	1
NDBI	0.0934	0.0747	30.2828	0.9433	0.7227	0.0747	0.0805	GAT-Temp Attn (MLP)	1	1
NDWI	0.0934	0.0747	30.2828	0.9433	0.7227	0.0747	0.0805	GAT-Temp Attn (MLP)	1	1
MNDWI	0.0934	0.0747	30.2828	0.9433	0.7227	0.0747	0.0805	GAT-Temp Attn (MLP)	1	1
EVI	0.2787	0.2673	183.7852	0.9359	-0.4322	0.2673	0.0829	GAT-Temp Attn (MLP)	1	1
SAVI	0.0816	0.0693	113.4220	0.9488	0.6509	0.0693	0.0485	GAT-Temp Attn (MLP)	1	1
MSAVI	0.0808	0.0699	123.0859	0.9455	0.6612	0.0699	0.0472	GAT-Temp Attn (MLP)	1	1
BSI	0.0911	0.0753	99.2389	0.9060	0.4174	0.0753	0.0578	GAT-Temp Attn (MLP)	1	1
GNDVI	0.0934	0.0747	30.2828	0.9433	0.7227	0.0747	0.0805	GAT-Temp Attn (MLP)	1	1
OSAVI	0.0912	0.0777	108.9220	0.9469	0.6304	0.0777	0.0518	GAT-Temp Attn (MLP)	1	1
NDVI	0.0571	0.0370	29.2285	0.9652	0.9305	0.0370	0.0571	LSTM (with aux)	-	1
NDBI	0.0449	0.0303	15.4073	0.9682	0.9360	0.0303	0.0449	LSTM (with aux)	-	1
NDWI	0.0449	0.0303	15.4073	0.9682	0.9360	0.0303	0.0449	LSTM (with aux)	-	1
MNDWI	0.0449	0.0303	15.4073	0.9682	0.9360	0.0303	0.0449	LSTM (with aux)	-	1
EVI	0.0700	0.0459	30.3401	0.9539	0.9096	0.0459	0.0700	LSTM (with aux)	-	1
SAVI	0.0375	0.0247	28.9224	0.9627	0.9263	0.0247	0.0375	LSTM (with aux)	-	1
MSAVI	0.0394	0.0260	29.9344	0.9592	0.9194	0.0260	0.0394	LSTM (with aux)	-	1
BSI	0.0344	0.0247	80.2508	0.9586	0.9169	0.0247	0.0344	LSTM (with aux)	-	1
GNDVI	0.0449	0.0303	15.4073	0.9682	0.9360	0.0303	0.0449	LSTM (with aux)	-	1
OSAVI	0.0395	0.0258	28.7094	0.9652	0.9309	0.0258	0.0395	LSTM (with aux)	-	1
NDVI	0.0775	0.0563	32.6249	0.9650	0.8721	0.0563	0.0730	LSTM (without aux)	-	1
NDBI	0.0837	0.0686	32.2712	0.9634	0.7772	0.0686	0.0655	LSTM (without aux)	-	1
NDWI	0.0837	0.0686	32.2712	0.9634	0.7772	0.0686	0.0655	LSTM (without aux)	-	1
MNDWI	0.0837	0.0686	32.2712	0.9634	0.7772	0.0686	0.0655	LSTM (without aux)	-	1
EVI	7.4517	0.2364	96.6464	0.0329	-	0.2364	7.4493	LSTM (without aux)	-	1
1022.5537										
SAVI	0.0415	0.0283	28.9869	0.9605	0.9095	0.0283	0.0414	LSTM (without aux)	-	1
MSAVI	0.0425	0.0291	30.0641	0.9582	0.9064	0.0291	0.0423	LSTM (without aux)	-	1
BSI	0.1115	0.1005	430.2640	0.9383	0.1271	0.1005	0.0567	LSTM (without aux)	-	1
GNDVI	0.0837	0.0686	32.2712	0.9634	0.7772	0.0686	0.0655	LSTM (without aux)	-	1
OSAVI	0.0474	0.0326	29.8677	0.9630	0.9002	0.0326	0.0469	LSTM (without aux)	-	1
NDVI	2.5350	0.1799	117.5658	0.0075	-135.9599	0.1799	2.5344	GAT-Temp Attn (MLP)(with aux)	1	1
NDBI	0.1181	0.0800	55.3663	0.8313	0.5567	0.0800	0.1001	GAT-Temp Attn (MLP)(with aux)	1	1
NDWI	0.1181	0.0800	55.3663	0.8313	0.5567	0.0800	0.1001	GAT-Temp Attn (MLP)(with aux)	1	1
MNDWI	0.1181	0.0800	55.3663	0.8313	0.5567	0.0800	0.1001	GAT-Temp Attn (MLP)(with aux)	1	1

EVI	0.2823	0.2425	177.4950	-0.5105	-0.4685	0.2425	0.2730	GAT-Temp Attn (MLP)(with aux)	1	1
SAVI	0.1326	0.1142	145.4538	0.5020	0.0772	0.1142	0.1281	GAT-Temp Attn (MLP)(with aux)	1	1
MSAVI	0.1361	0.1155	160.9344	0.4599	0.0387	0.1155	0.1298	GAT-Temp Attn (MLP)(with aux)	1	1
BSI	0.0772	0.0507	379.3436	0.8201	0.5820	0.0507	0.0687	GAT-Temp Attn (MLP)(with aux)	1	1
GNDVI	0.1181	0.0800	55.3663	0.8313	0.5567	0.0800	0.1001	GAT-Temp Attn (MLP)(with aux)	1	1
OSAVI	0.1427	0.1227	133.2936	0.5952	0.0957	0.1227	0.1378	GAT-Temp Attn (MLP)(with aux)	1	1
NDVI	0.1308	0.1026	106.1529	0.8570	0.6352	0.1026	0.1296	GAT-Temp Attn (MLP)	2	1
NDBI	0.1268	0.1042	53.0599	0.9238	0.4886	0.1042	0.1244	GAT-Temp Attn (MLP)	2	1
NDWI	0.1268	0.1042	53.0599	0.9238	0.4886	0.1042	0.1244	GAT-Temp Attn (MLP)	2	1
MNDWI	0.1268	0.1042	53.0599	0.9238	0.4886	0.1042	0.1244	GAT-Temp Attn (MLP)	2	1
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MSAVI	0.0949	0.0826	149.2168	0.9244	0.5324	0.0826	0.0947	GAT-Temp Attn (MLP)	2	1
BSI	0.1166	0.0928	382.0753	0.6772	0.0462	0.0928	0.0998	GAT-Temp Attn (MLP)	2	1
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NDVI	0.1424	0.1176	101.9778	0.9234	0.5677	0.1176	0.0839	GAT-Temp Attn (MLP)	1	1
NDBI	0.0934	0.0747	30.2828	0.9433	0.7227	0.0747	0.0805	GAT-Temp Attn (MLP)	1	1
NDWI	0.0934	0.0747	30.2828	0.9433	0.7227	0.0747	0.0805	GAT-Temp Attn (MLP)	1	1
MNDWI	0.0934	0.0747	30.2828	0.9433	0.7227	0.0747	0.0805	GAT-Temp Attn (MLP)	1	1
EVI	0.2787	0.2673	183.7852	0.9359	-0.4322	0.2673	0.0829	GAT-Temp Attn (MLP)	1	1
SAVI	0.0816	0.0693	113.4220	0.9488	0.6509	0.0693	0.0485	GAT-Temp Attn (MLP)	1	1
MSAVI	0.0808	0.0699	123.0859	0.9455	0.6612	0.0699	0.0472	GAT-Temp Attn (MLP)	1	1
BSI	0.0911	0.0753	99.2389	0.9060	0.4174	0.0753	0.0578	GAT-Temp Attn (MLP)	1	1
GNDVI	0.0934	0.0747	30.2828	0.9433	0.7227	0.0747	0.0805	GAT-Temp Attn (MLP)	1	1
OSAVI	0.0912	0.0777	108.9220	0.9469	0.6304	0.0777	0.0518	GAT-Temp Attn (MLP)	1	1